

HAIG-SIMONS INCOME^{*}

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Abstract

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1 Theoretical Framework

Aggregate output Y_t is produced by a representative firm with production function $Y_t = F(K_t, L_t)$, where K_t is capital and L_t is labor. Firm profit equals $\Pi_t = F(K_t, L_t) - w_t L_t - \rho_t K_t$, where w_t is the wage rate, and ρ_t the rental rate of capital.¹ All assets are owned by the business sector, and business income (BI_t) includes both (implicit) capital rental income as well as profits: $BI_t \equiv \Pi_t + \rho_t K_t - \delta K_t$. Out of this income, firms either pay dividends d_t or retain the earnings $RE_t = BI_t - d_t$.

Capital evolves through the law of motion $K_{t+1} = (1 - \delta)K_t + I_t$. Investment is the sum of retained earnings (gross of depreciation) and net equity issuances, NE_t , which equals share issuances minus buybacks. We then have $I_t = RE_t + \delta K_t + NE_t$.

The market value of the firm (share price P_t times shares outstanding S_t) equals the value of its capital plus the present discounted value of future profits, denoted by V_t .

$$P_t S_t = K_{t+1} + E_t \left[\sum_{s=1}^{\infty} M_{t,t+s} \Pi_{t+s} \right] \equiv K_{t+1} + V_t, \quad (1)$$

where $M_{t,t+s}$ is the stochastic discount factor. Implicit in this equation is the Miller and Modigliani [1961] assumption that a dollar of retained earnings increases the value of the firm by a dollar.

We furthermore assume that equity issuances and buybacks are at the prevailing market price, $NE_t = P_t(S_t - S_{t-1})$. This ensures that changes in net equity have no effect on the price per share.²

There are two distinct types of income present in this economy.

Definition 1. (*Net of depreciation*) *National Income is equal to income from wages plus dividends plus retained earnings.*

$$Y_t^n = w_t L_t + d_t + RE_t \quad (2)$$

National income is income received from production. Definition 1 is in line with the BEA definition of national income, which, absent measurement errors, equals the net national product: $Y_t^n = Y_t - \delta K_t$.³

Definition 2. *Capital gains equal the real increase in the price per share of equity times the value of equity. Defining $r_t^p = P_t/P_{t-1} - 1$, $KG_t = P_{t-1}S_{t-1} \cdot r_t^p$.*

Capital gains are the increase in wealth caused by changes in asset prices. From Equation 1, the law of motion for capital, and the share repurchase equation, we

¹These are thus ‘pure’ or factorless profits, as in Karabarbounis and Neiman [2019].

²To see this, define shareholder value as $W = PS$ and consider a deviation in net equity raised Δ , at a price per share of P . Then new firm value is $W' = W + \Delta$, and new shares outstanding is $S' = S + \Delta/P$. The price per share is then $W'/S' = (PS + \Delta)/(S + \Delta/P) = P$.

³See the BEA Handbook, Fox and McCully [2009].

can decompose gains into corporate retained earnings and changes in the value of future discounted profits:

$$KG_t = \Delta V_t + RE_t. \quad (3)$$

The BEA definition of national income explicitly excludes capital gains, which are not earned in production. However, as shown in Equation 3, capital gains are indirectly present through retained earnings. To isolate capital gains from other income flows, we define ordinary factor income as national income minus the retained earnings of corporations: $Y_t^o = Y_t^n - RE_t$, and *pure* capital gains as capital gains minus retained earnings, $KG_t^p = KG_t - RE_t$.

To measure aggregate capital gains in the data, we use data from the Financial Accounts of the Federal Reserve, which collects information on the balance sheet and asset revaluations of the U.S. household sector. We convert nominal to real gains by subtracting net national product inflation, and separate total from pure capital gains by subtracting the BEA series for corporate savings.

Figure 2 displays the results. Capital gains are large, especially in the post-1980 period, where they average 14% of national income. Pure capital gains are the majority, averaging 10% of national income, while retained earnings, the component present in national income, averaged 4%. The large magnitude and persistence of capital gains suggests that analyses that do not include this income source provide an incomplete picture of inequality in recent times.

The growth in the economic importance of capital gains motivates a comprehensive income definition that includes changes in asset prices as well as standard income components.

Definition 3. *Haig-Simons income [Haig, 1921, Simons, 1938] equals ordinary national income plus capital gains: $HS_t = Y_t^o + KG_t$.*

Compared with other income definitions, Haig-Simons has several desirable properties.

Remark 1. *The change in household wealth, $W_t = P_t S_t$, between two periods is equal to Haig-Simons income (on a post-tax post-transfer basis) minus consumption: $\Delta W_t = HS_t - c_t$.*

To see this, we introduce a household sector that holds all assets and receives all income, with flow budget constraint given by

$$c_t + P_t S_t = S_{t-1} P_t + d_t + w_t L_t \quad (4)$$

Subtracting W_{t-1} yields $\Delta W_t = (Y_t^n - c_t) + KG_t^p$, i.e. national savings plus pure gains. Haig-Simons income can thus more accurately account for changes in wealth compared to national income and national saving.

Capital gains are also closely related to consumer welfare.

Remark 2. *From the household's budget constraint, capital gains and other capital income sources are interchangeable: a dollar of gains expands the agent's feasible consumption set identically to a dollar of interest or dividends.*

In Appendix J we formally show that holding expected future returns constant, a dollar of capital gains increases welfare by the marginal utility of consumption. For this reason, when studying relative inequality, Haig-Simons is preferred to other income concepts since it better distinguishes welfare not only between asset holders and non-asset holders, but between investors who hold different types of assets. Consider two investors with identical wealth invested in assets earning the same total return. Investor A holds a bond paying a 5% coupon; Investor B holds a growth stock that pays no dividend but appreciates 5% per year. Under national income, Investor A earns \$5 per \$100 invested, while Investor B earns zero — despite being economically identical. More broadly, any income measure that excludes capital gains will systematically understate the income of growth stock holders relative to dividend stock holders, equity holders relative to bond holders, and business founders relative to salaried executives.

What are the sources of aggregate capital gains, and why have they grown over time?

Remark 3. *From Equations 1 and 3, capital gains may be driven by (i) changes in firm profits, (ii) changes in retained earnings, or (iii) changes in discount rates.*

In the U.S. context, there is substantial evidence that all three channels have played a role in the rise in capital gains. For (i), Grullon et al. [2019], Autor et al. [2020] document a secular increase in concentration and market power, which in turn has led to increases in both earnings and profits [Barkai, 2020, Eeckhout, 2024].

For (ii), a large literature has documented a secular increase in intangible investment in the U.S.,⁴ which in turn has been capitalized into higher stock market valuations [Hall, 2001, McGrattan and Prescott, 2010]. These investments, however, are often unrecorded under GAAP and national income accounting standards [Lev, 2001, Corrado et al., 2009].⁵ As a result, measured corporate profits and retained earnings are lower than their true values. If intangible investment is undermeasured in the national accounts but observed and valued by asset markets, it would show up in our measurement as an increase in KG_t^p .

Outside of these secular shifts, standard trend-growth also generates capital gains. Under a balanced growth path in which all variables grow at rate g , asset prices will also follow this trend, and the ratio of capital gains to GDP will be $g \cdot \frac{W}{Y}$. Plugging in the value of public and private equities held by US households for W and a real GDP growth rate of 2% yields a back-of-the-envelope estimation of 5% of national income in capital gains per year.

For (iii), there has been a decline in real interest rates and estimates of the natural rate of interest in the United States [Eggertsson et al., 2019, Laubach and

⁴See, for example, [Crouzet et al., 2022, Peters and Taylor, 2017].

⁵Although the BEA has revised its treatment of intangible investment to include additional categories such as software, it still excludes firm-specific human capital, organization capital, brand equity, and advertising.

Williams, 2016] and globally [Rachel and Summers, 2019]. Changes in discount rates complicate the interpretation of Haig-Simons income because in addition to the standard wealth effect of higher asset prices, there is a separate welfare effect of lower future returns. We discuss the impact of lower interest rates (and more generally, changes in discount rates) in Section J.

Remark 4. *Changes in firm payout policy can affect total capital gains KG_t , but not national income or pure gains KG_t^p . As a result, Haig-Simons income is payout irrelevant.*

From Equation 3, changes in stock buybacks *holding dividends fixed* have no effect on KG_t , since RE_t is unaffected. If firms substitute buybacks for dividends, however, RE_t and KG_t increase. In contrast, both pure gains KG_t^p and national income are unaffected, since KG_t^p excludes retained earnings and national income is pinned down by the production side of the economy. Since its two subcomponents are payout irrelevant, Haig-Simons income is as well.

Remark 5. *The Haig-Simons concept does not double count income.*

Canberra [2001], Splinter [2025] argue that Haig-Simons double counts income, since higher expected future earnings $E_t\Pi_{t+s}$ count once initially when stock prices increase (higher ΔV_t), and again when earnings are paid out (higher d_{t+s}). This analysis, however, overlooks the fact that paying out dividends reduces capital gains one-to-one with distributions; this is exactly the reason why stock prices fall on ex-dividend dates. The fact that future dividend payments subtract from capital gains ensures that the increase in future earnings is only counted once.⁶

References

- David Autor, David Dorn, Lawrence F Katz, Christina Patterson, and John Van Reenen. The fall of the labor share and the rise of superstar firms. *The Quarterly journal of economics*, 135(2):645–709, 2020.
- Simcha Barkai. Declining labor and capital shares. *The Journal of Finance*, 75(5):2421–2463, 2020.
- Canberra. Final report and recommendations. Technical report, The Canberra Group, Ottawa, 2001.
- Carol Corrado, Charles Hulten, and Daniel Sichel. Intangible capital and us economic growth. *Review of income and wealth*, 55(3):661–685, 2009.
- Nicolas Crouzet, Janice C Eberly, Andrea L Eisfeldt, and Dimitris Papanikolaou. The economics of intangible capital. *Journal of Economic Perspectives*, 36(3): 29–52, 2022.

⁶A similar argument to this was made in Larrimore et al. [2021].

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- Jan Eeckhout. The value and profits of firms. Technical report, CEPR Discussion Papers, 2024.
- Gauti B Eggertsson, Neil R Mehrotra, and Jacob A Robbins. A model of secular stagnation: Theory and quantitative evaluation. *American Economic Journal: Macroeconomics*, 11(1):1–48, 2019.
- Douglas R Fox and Clinton P McCully. Concepts and methods of the us national income and product accounts. *NIPA Handbook*, 2009.
- Gustavo Grullon, Yelena Larkin, and Roni Michaely. Are us industries becoming more concentrated? *Review of Finance*, 23(4):697–743, 2019.
- Robert M Haig. The concept of income-economic and legal aspects. *The federal income tax*, 1(7), 1921.
- Robert E Hall. The stock market and capital accumulation. *American Economic Review*, 91(5):1185–1202, 2001.
- Loukas Karabarbounis and Brent Neiman. Accounting for factorless income. *NBER Macroeconomics Annual*, 33(1):167–228, 2019.
- Jeff Larrimore, Richard V Burkhauser, Gerald Auten, and Philip Armour. Recent trends in us income distributions in tax record data using more comprehensive measures of income including real accrued capital gains. *Journal of Political Economy*, 129(5):1319–1360, 2021.
- Thomas Laubach and John C Williams. Measuring the natural rate of interest redux. *Business Economics*, 51(2):57–67, 2016.
- Baruch Lev. *Intangibles: Management, Measurement, and Reporting*. Brookings Institution Press, Washington, DC, 2001.
- Ellen R McGrattan and Edward C Prescott. Unmeasured investment and the puzzling us boom in the 1990s. *American Economic Journal: Macroeconomics*, 2(4):88–123, 2010.
- Merton H Miller and Franco Modigliani. Dividend policy, growth, and the valuation of shares. *the Journal of Business*, 34(4):411–433, 1961.
- Ryan H Peters and Lucian A Taylor. Intangible capital and the investment-q relation. *Journal of Financial Economics*, 123(2):251–272, 2017.
- Łukasz Rachel and Lawrence H Summers. On secular stagnation in the industrialized world. Technical report, National Bureau of Economic Research, 2019.
- Henry C Simons. *Personal Income Taxation: The Definition of Income as a Problem of Fiscal Policy*. University of Chicago Press, Chicago, 1938.

David Splinter. The distribution of capital gains in the united states: A comment.
2025.